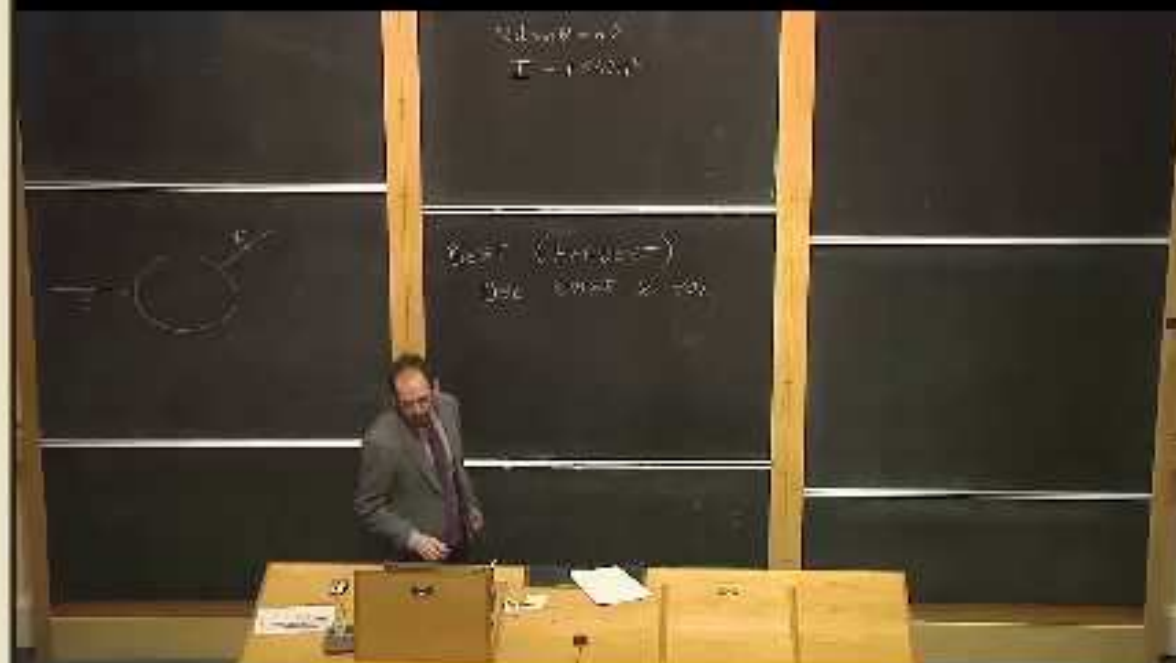
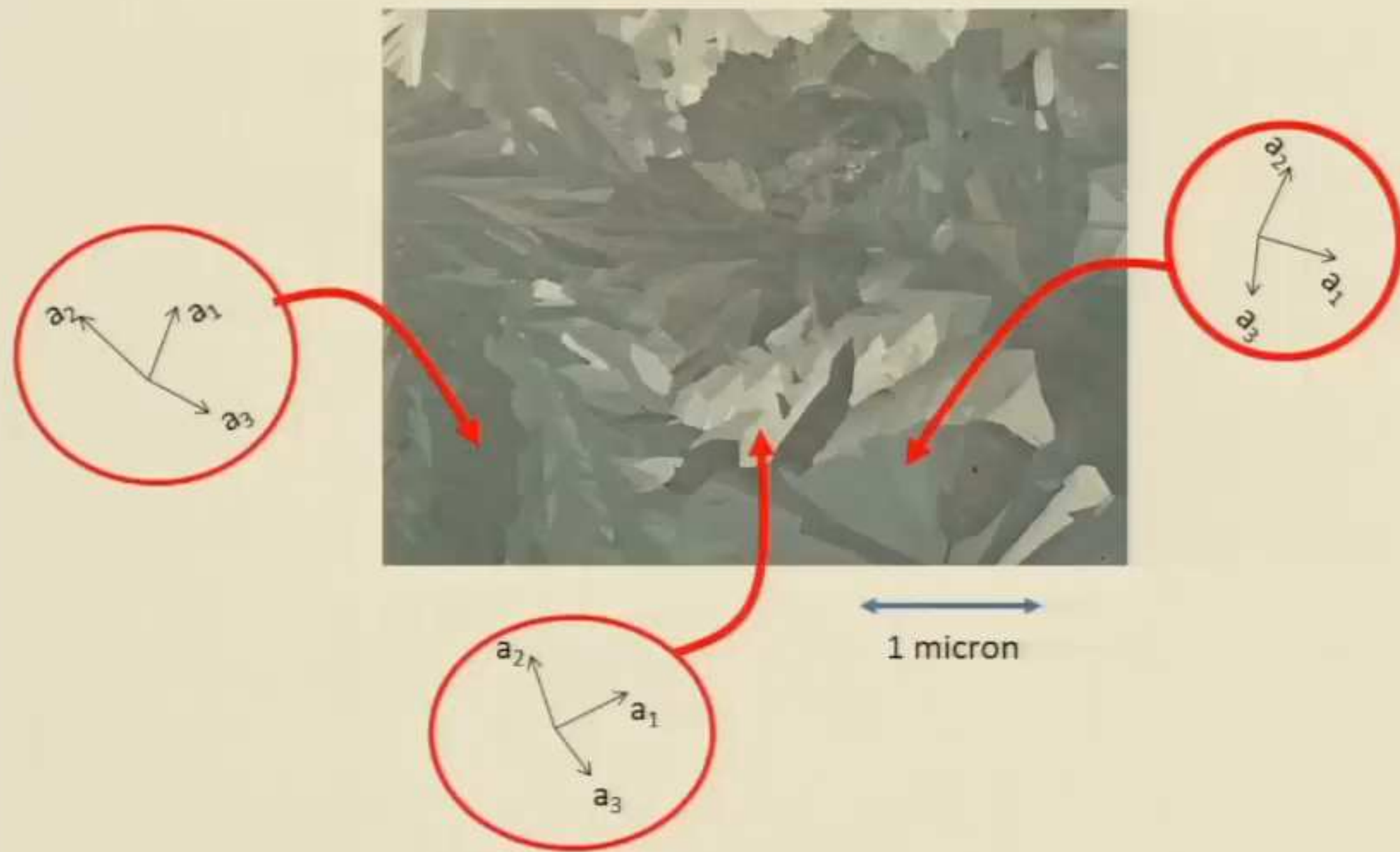


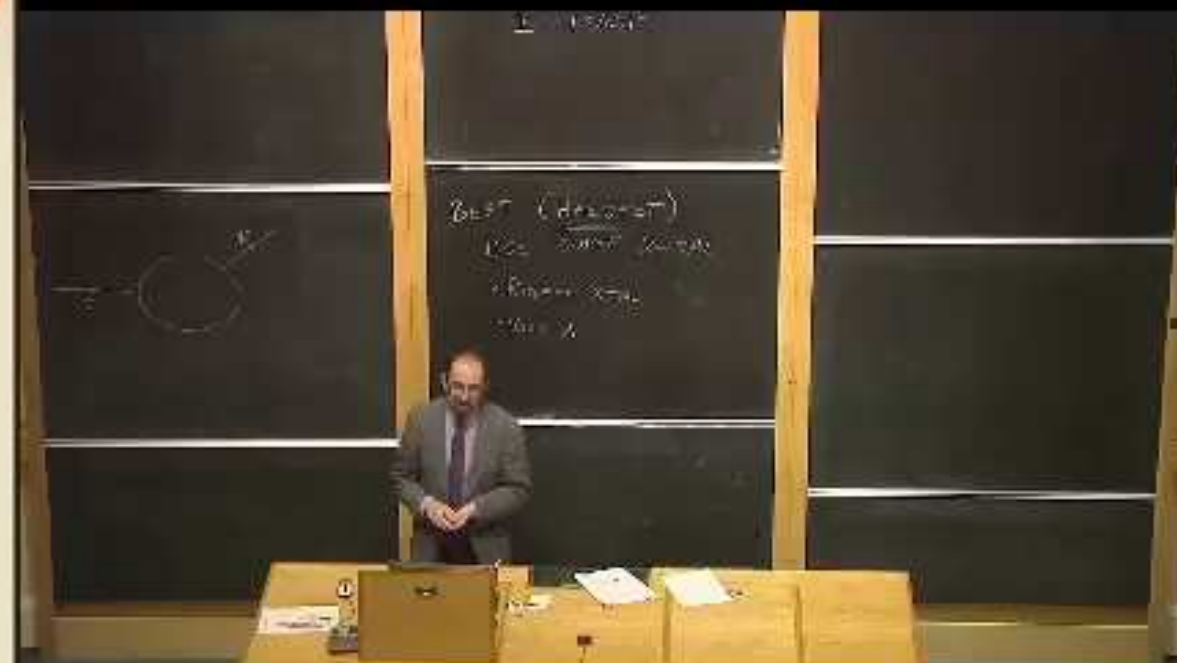


Nice single crystals (NaCl)





Polycrystalline Material





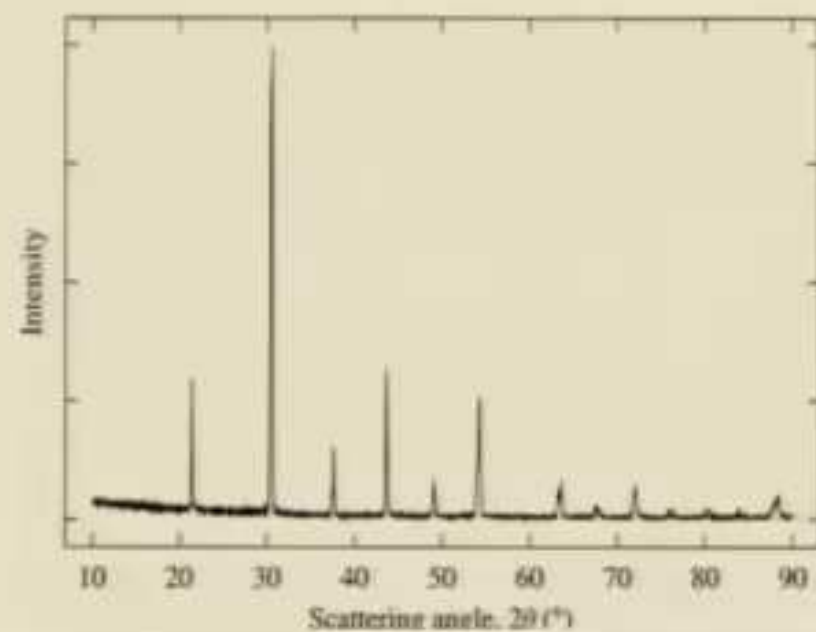
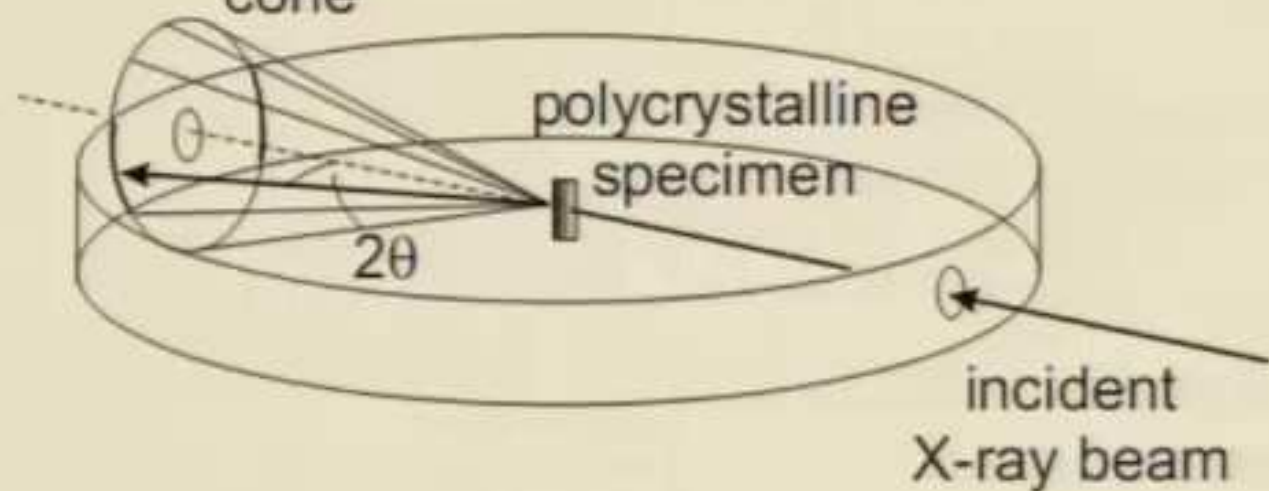
X-ray diffraction rings recorded on a photographic film

$2\theta = 0^\circ$

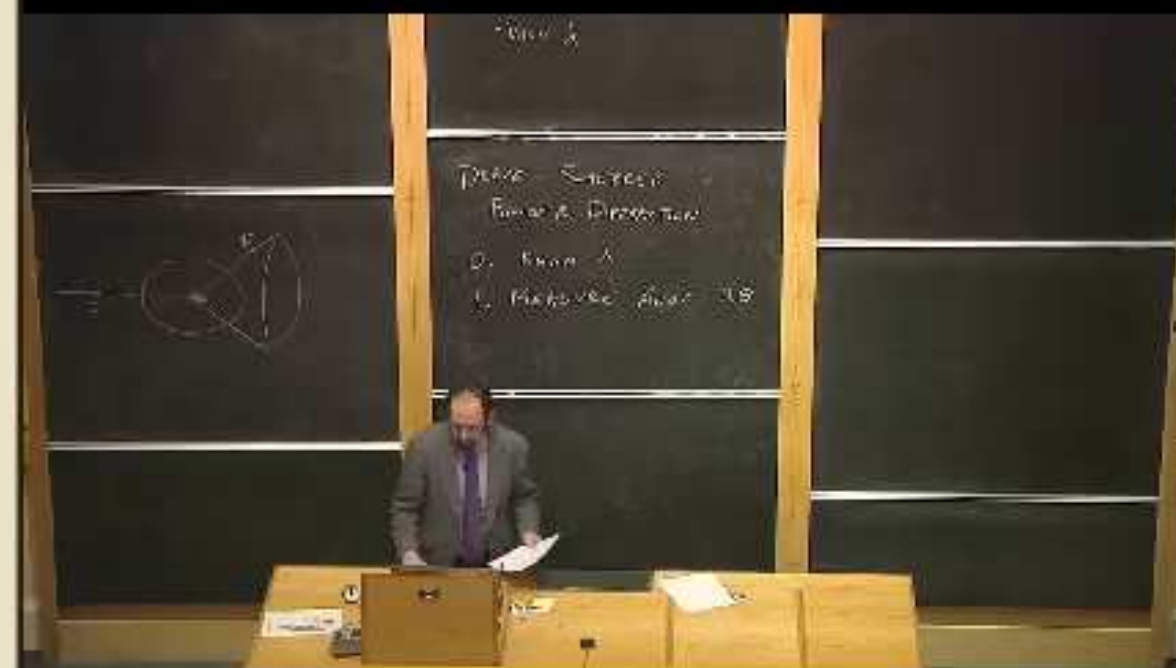
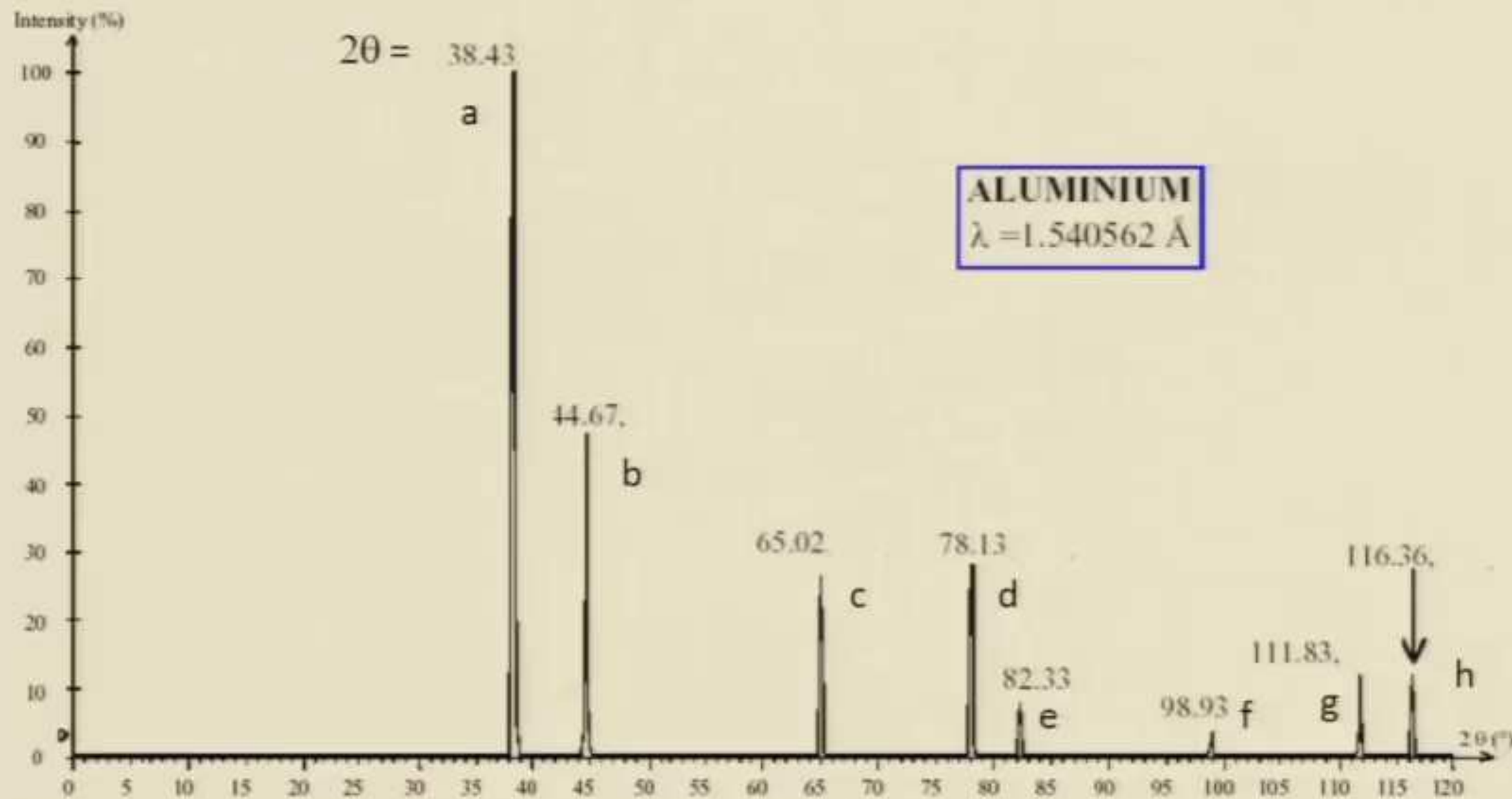
$2\theta = 90^\circ$

$2\theta = 180^\circ$

Debye-Scherrer cone



Consider the following XRD pattern for Aluminum, which was collected using $\text{CuK}\alpha$ radiation.



Scattering Selection Rules

P = Primitive (simple) cubic

I = BCC

F = FCC

All hkl

$h+k+l = \text{even}$

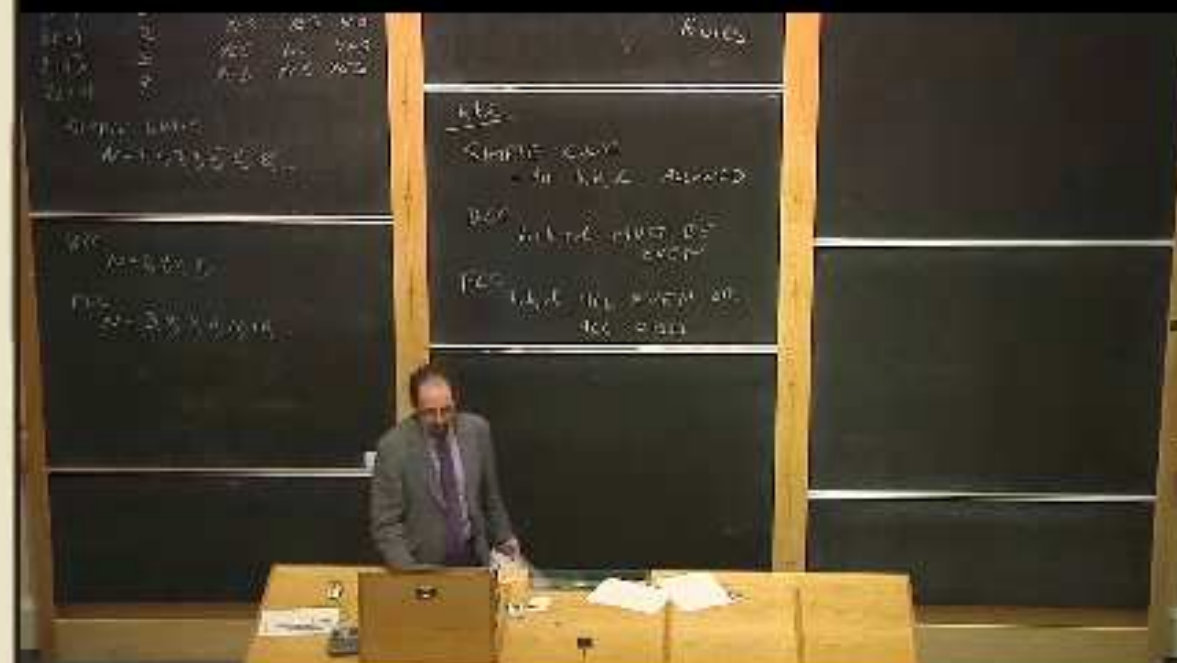
h, k, l all even or all odd

$\{hkl\}$	$N=h^2+k^2+l^2$	P	I	F
100	1	*		
110	2	*	*	
111	3	*		*
200	4	*	*	*
210	5	*		
211	6	*	*	
---	7			
220	8	*	*	*
221, 300	9	*		
310	10	*	*	
311	11	*		*
222	12	*	*	*
320	13	*		
321	14	*	*	
---	15			
400	16	*	*	*



Peak Angle 2θ

a	38.43
b	44.67
c	65.02
d	78.13
e	82.33
f	98.93
g	111.83
h	116.36



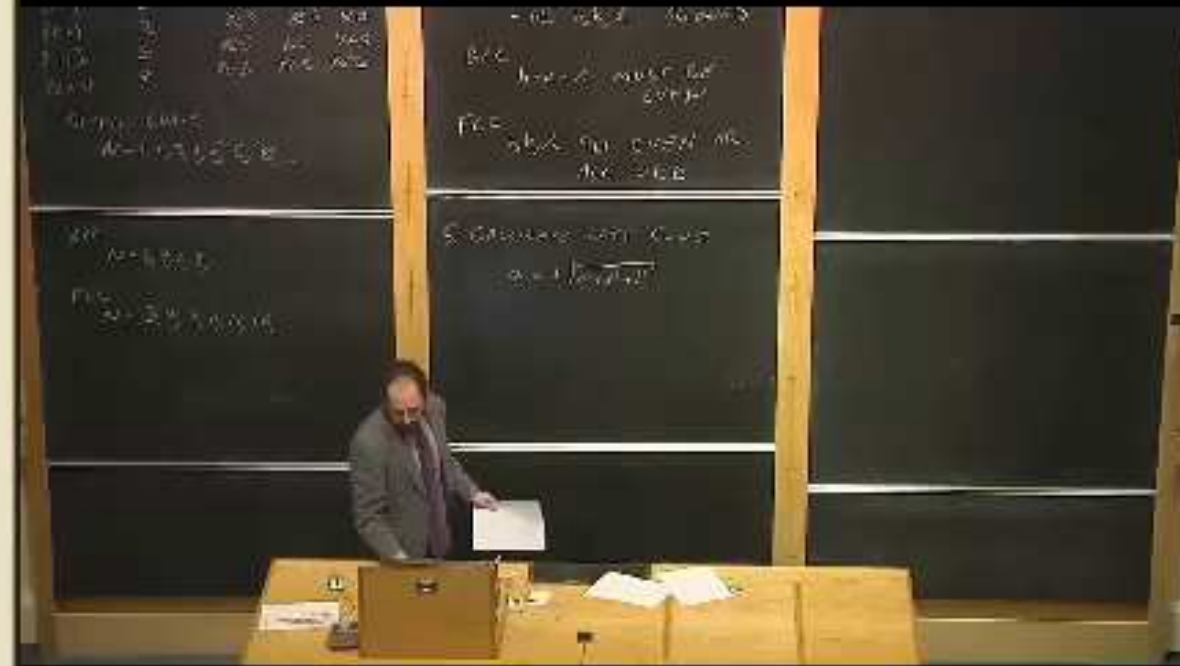
$$a^2/d^2 = h^2 + k^2 + l^2$$

$$N = h^2 + k^2 + l^2$$

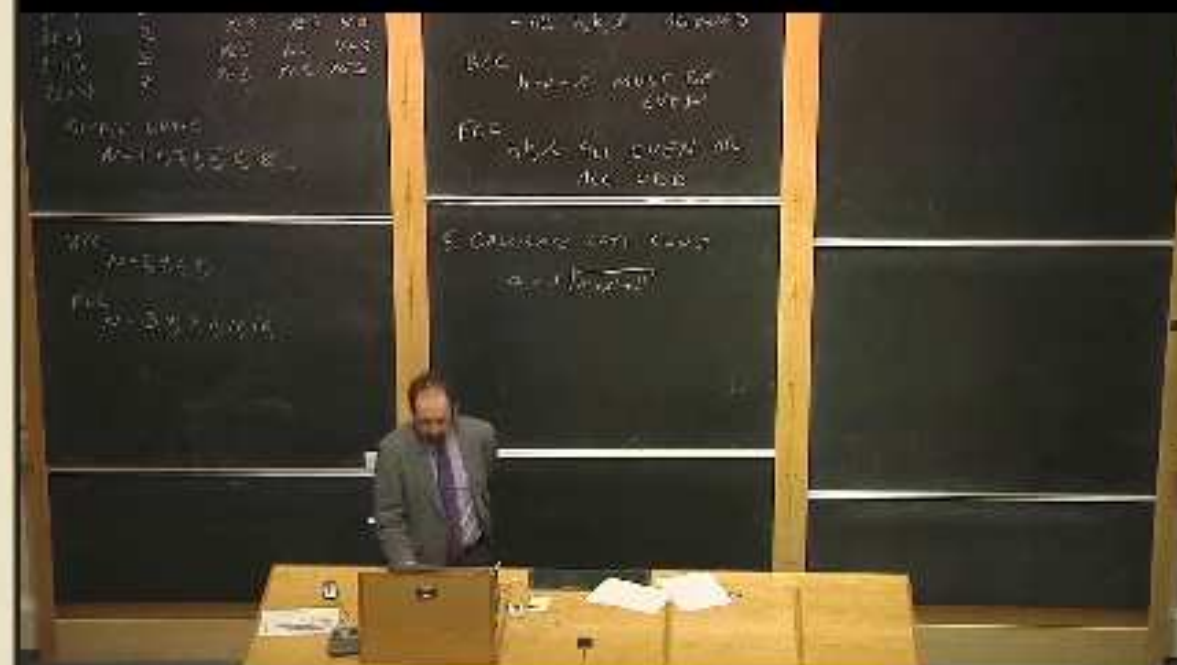
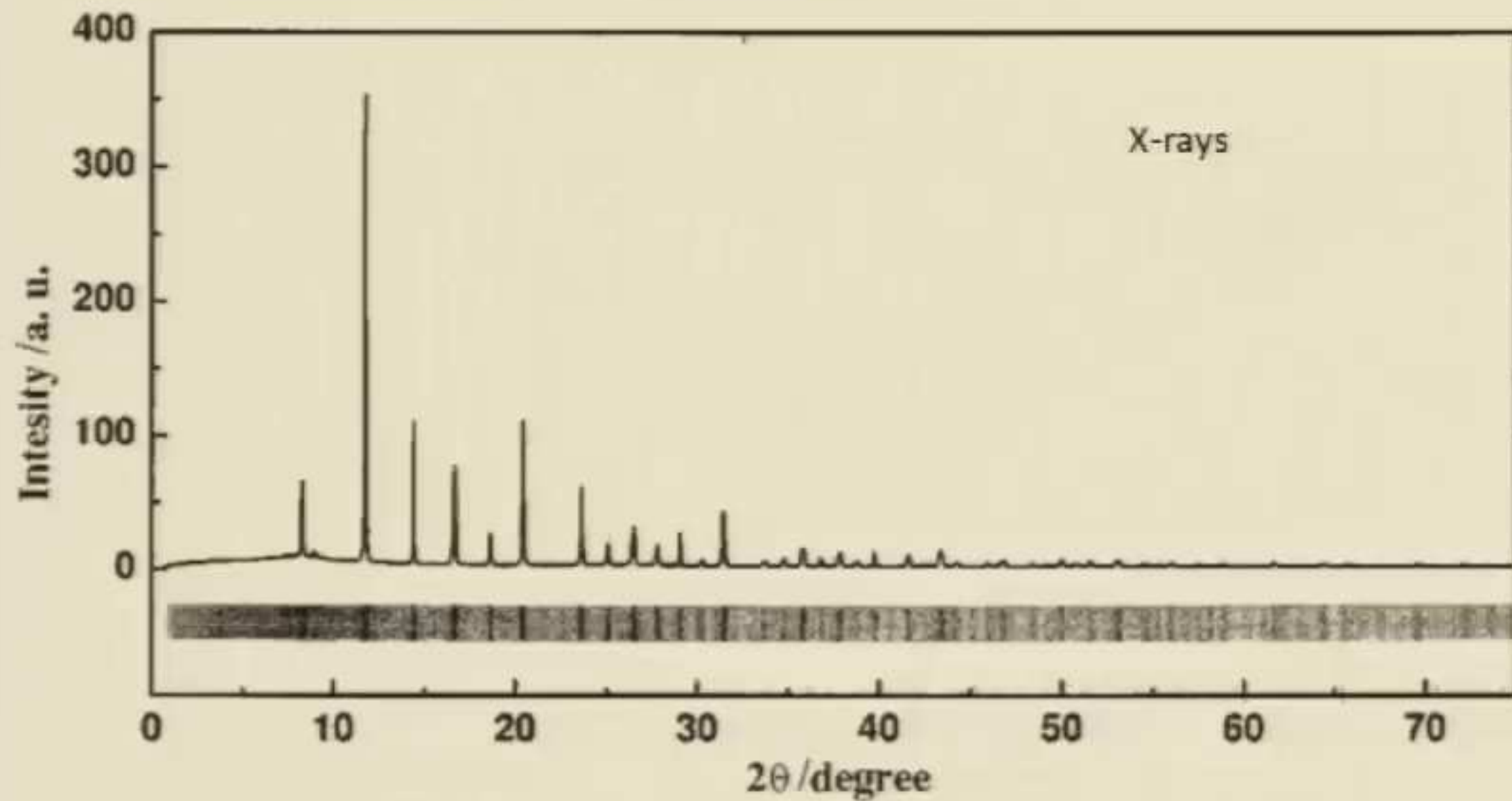


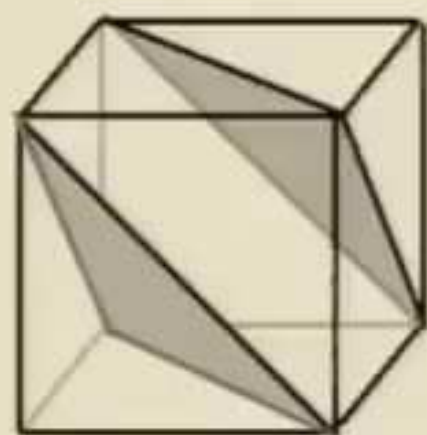
Peak	Angle 2θ	$d = \frac{\lambda}{2 \sin \theta}$	d_a^2/d^2	$3d^2/d_a^2$	N
a	38.43	2.3405Å	1.0000	3.0000	3
b	44.67	2.0269Å	1.3333	3.9999	4
c	65.02	1.4332Å	2.6667	8.0002	8
d	78.13	1.2223Å	3.6666	10.9999	11
e	82.33	1.1702Å	4.0000	12.0001	12
f	98.93	1.0135Å	5.3327	15.9980	16
g	111.83	0.9301Å	6.3327	18.9980	19
h	116.36	0.9065Å	6.6657	19.9972	20

FCC! : h,k,l all even or all odd : $N = 3,4,8,11,12 \dots$

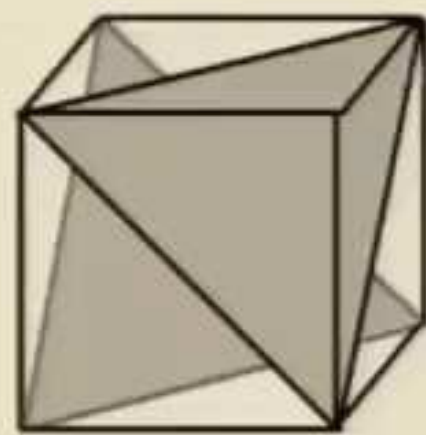


Barium Titanate

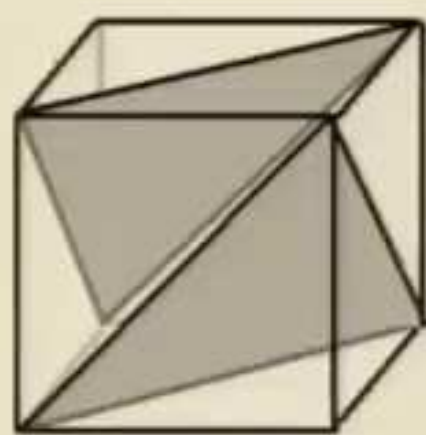




(111)



$(1\bar{1}1)$



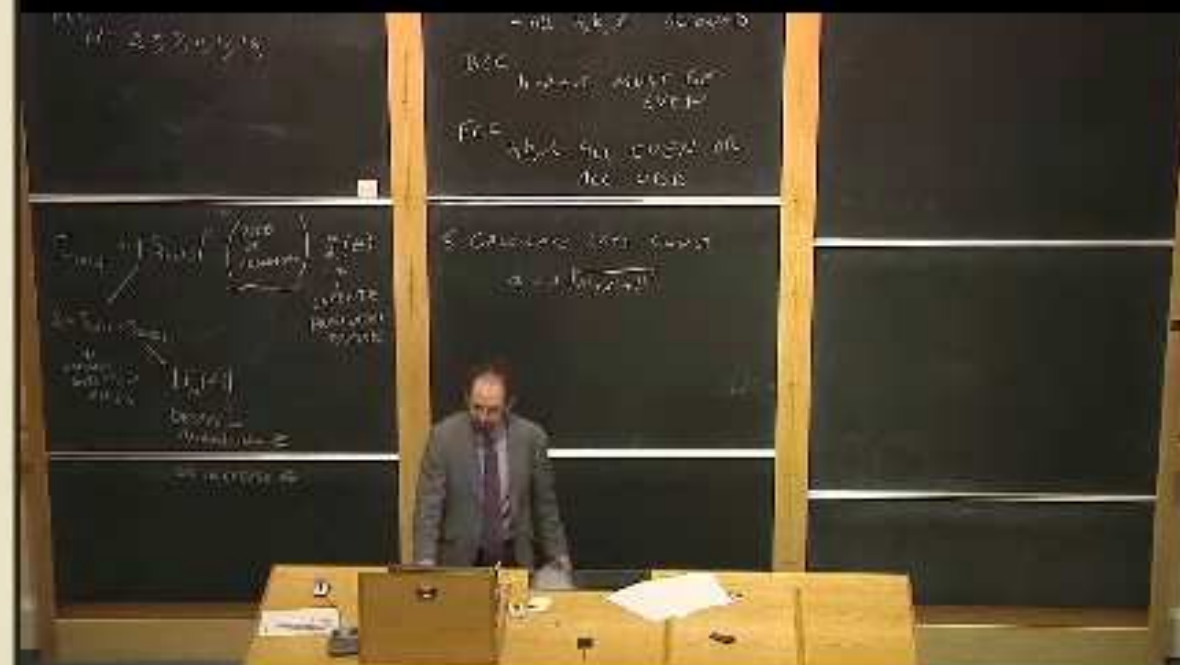
$(\bar{1}11)$

Three of the 8 possible families of lattice planes in the $\{111\}$ class.

All of these faces show the same spacing between lattice planes.

In a powder experiment all of these scatter at the same angle

Intensity of scattering is weighted by the multiplicity



Scattering Selection Rules

P = Primitive (simple) cubic

I = BCC

F = FCC

All hkl

$h+k+l = \text{even}$

h, k, l all even or all odd

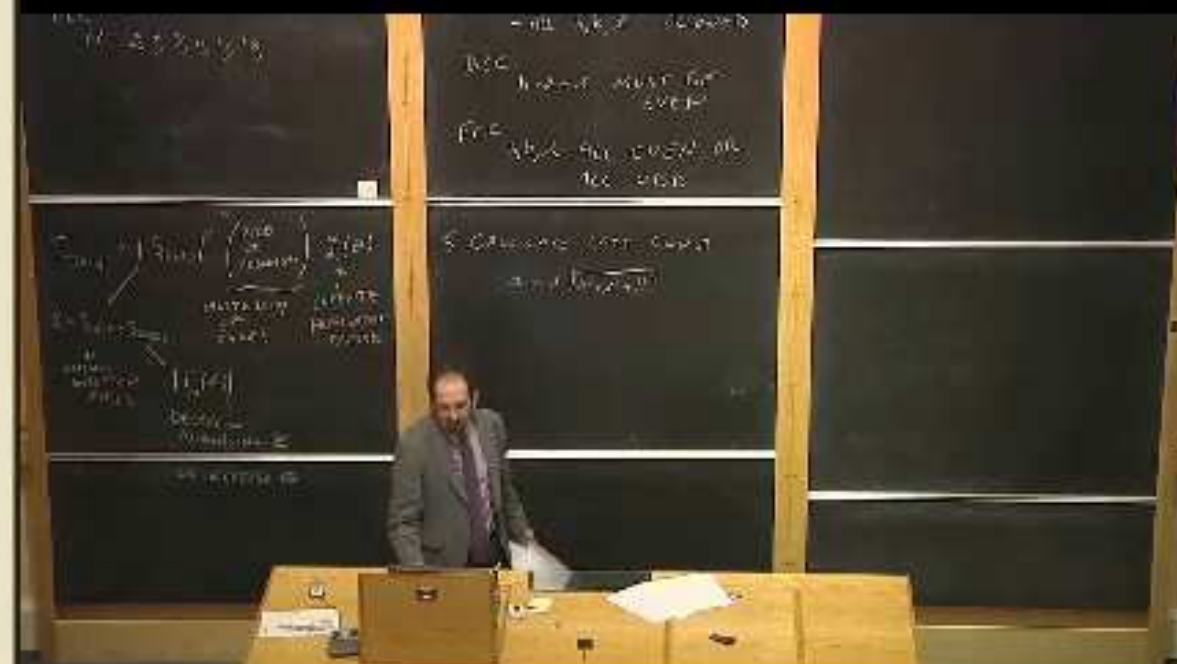
$\{hkl\}$	$N=h^2+k^2+l^2$	Multiplicity	P	I	F
100	1	6	*		
110	2	12	*	*	
111	3	8	*		*
200	4	6	*	*	*
210	5	24	*		
211	6	24	*	*	
---	7	--			
220	8	12	*	*	*
221, 300	9	24+6	*		
310	10	24	*	*	
311	11	24	*		*
222	12	8	*	*	*
320	13	24	*		
321	14	48	*	*	
---	15	--			
400	16	6	*	*	*

Sequence of
N values

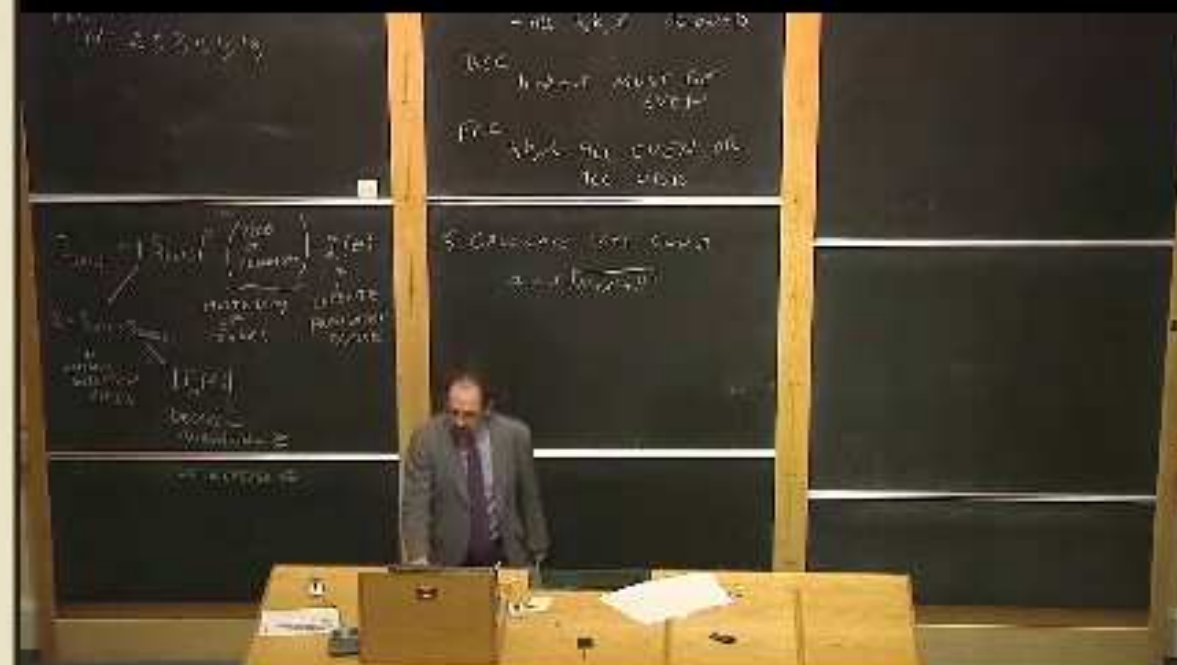
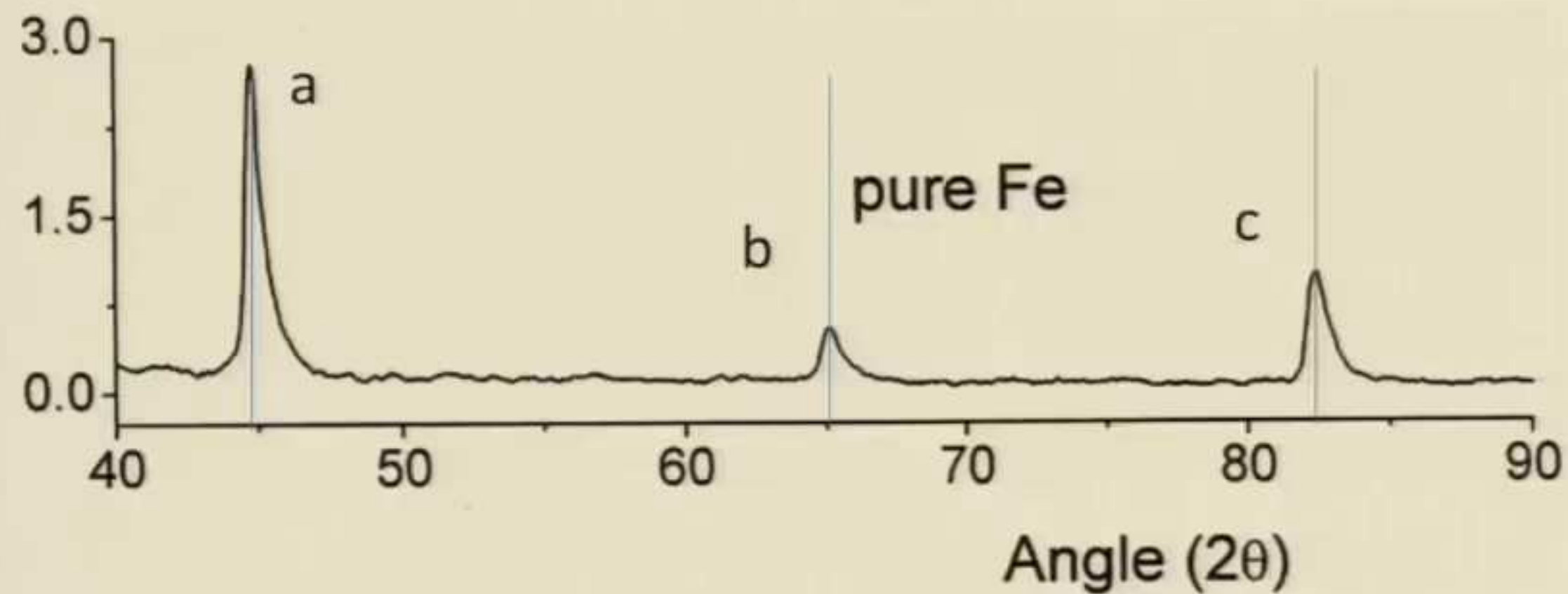
P: 1, 2, 3, 4, 5, 6, 8, 9, (= all integers excluding 7, 15, 23, ...)

I: 2, 4, 6, 8, 10, 12, 14, ... (= even integers excluding 28, 60, ...)

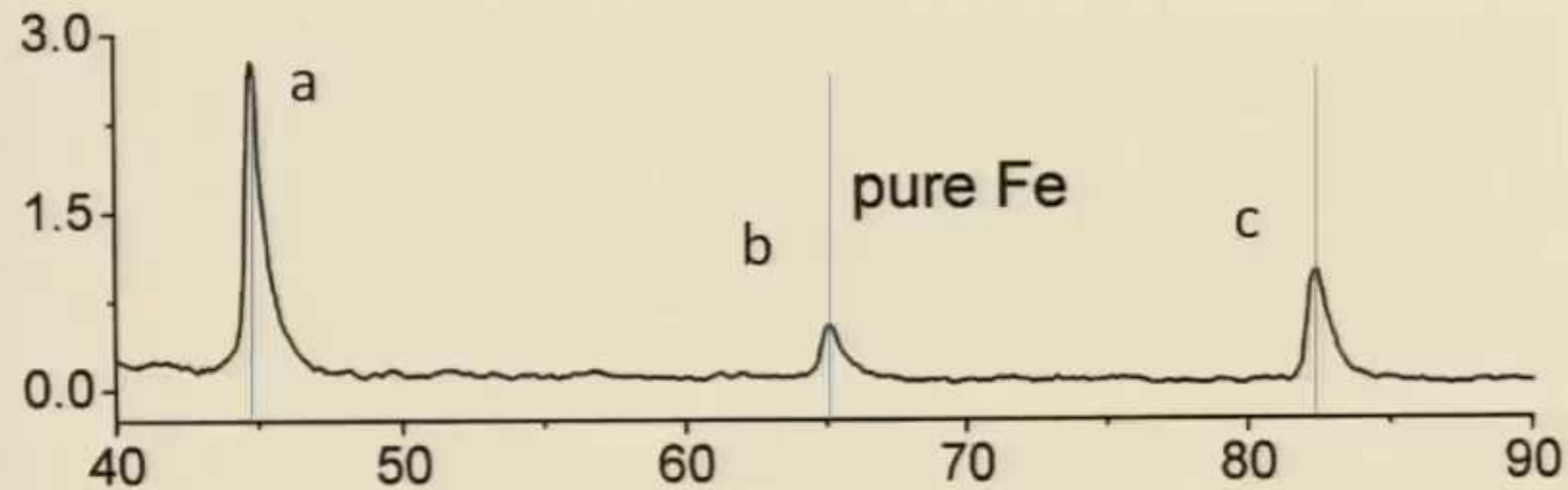
F: 3, 4, 8, 11, 12, 16, 19, 20, ...



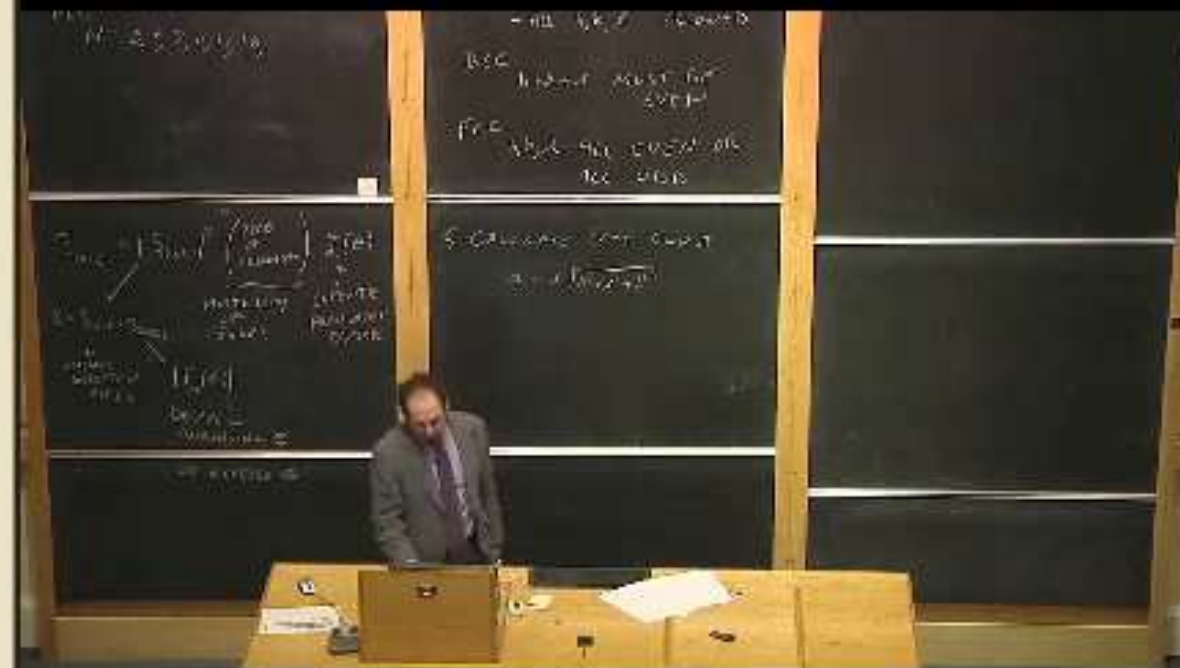
X-ray $\lambda=1.54$ Angstrom



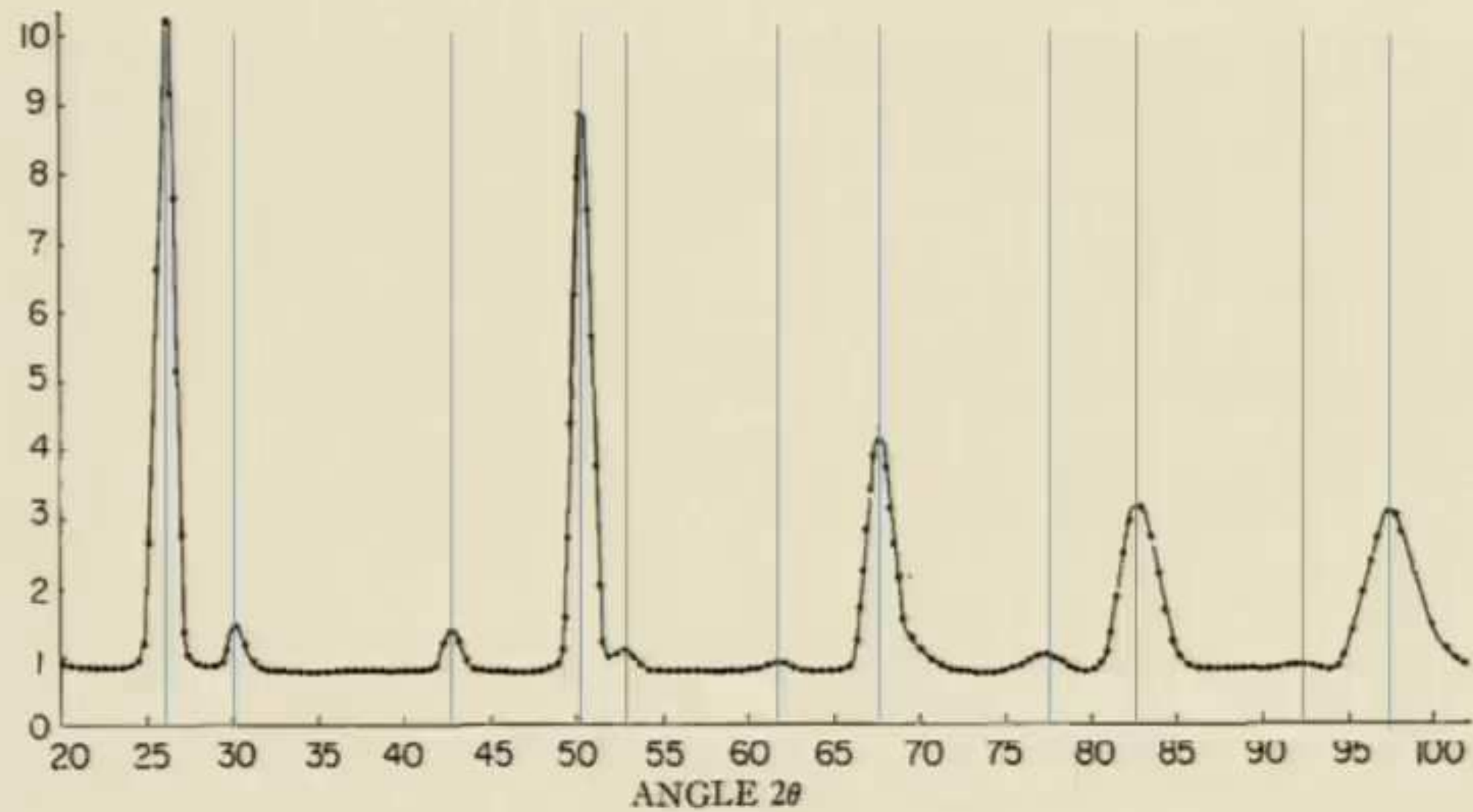
X-ray $\lambda=1.54$ Angstrom



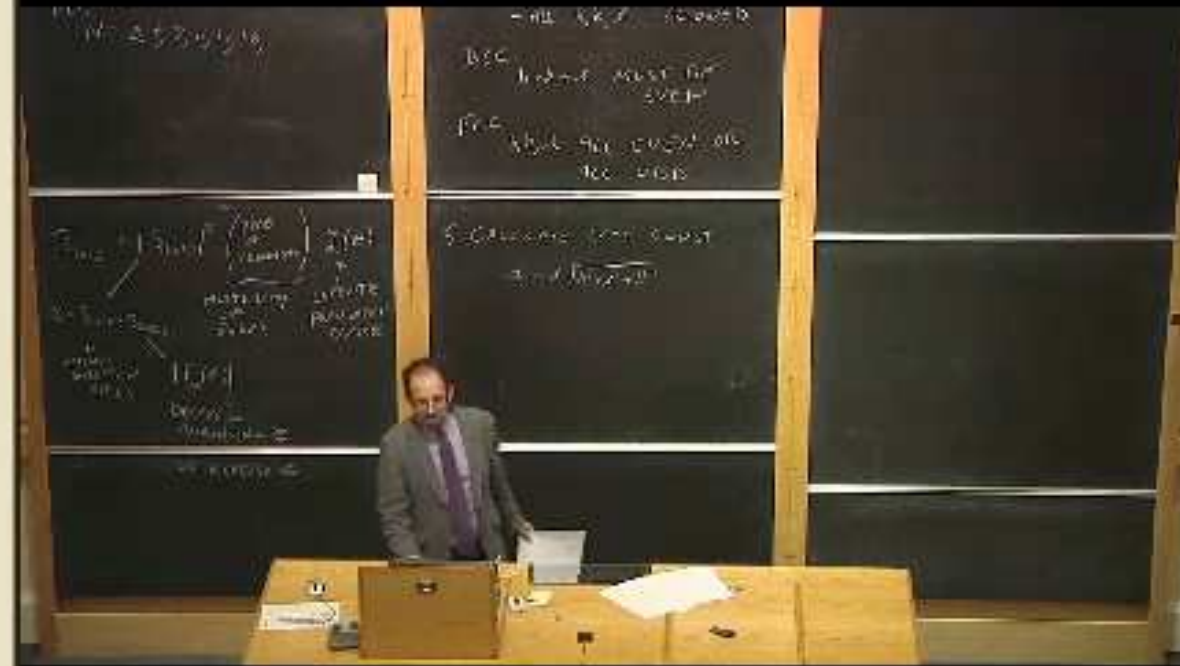
Simple Cubic	{100}	{110}	{111}
Multiplicity	6	12	8
BCC	{110}	{200}	{211}
Multiplicity	12	6	24



$\lambda = 1.09$ Angstrom TiC neutron powder diffraction



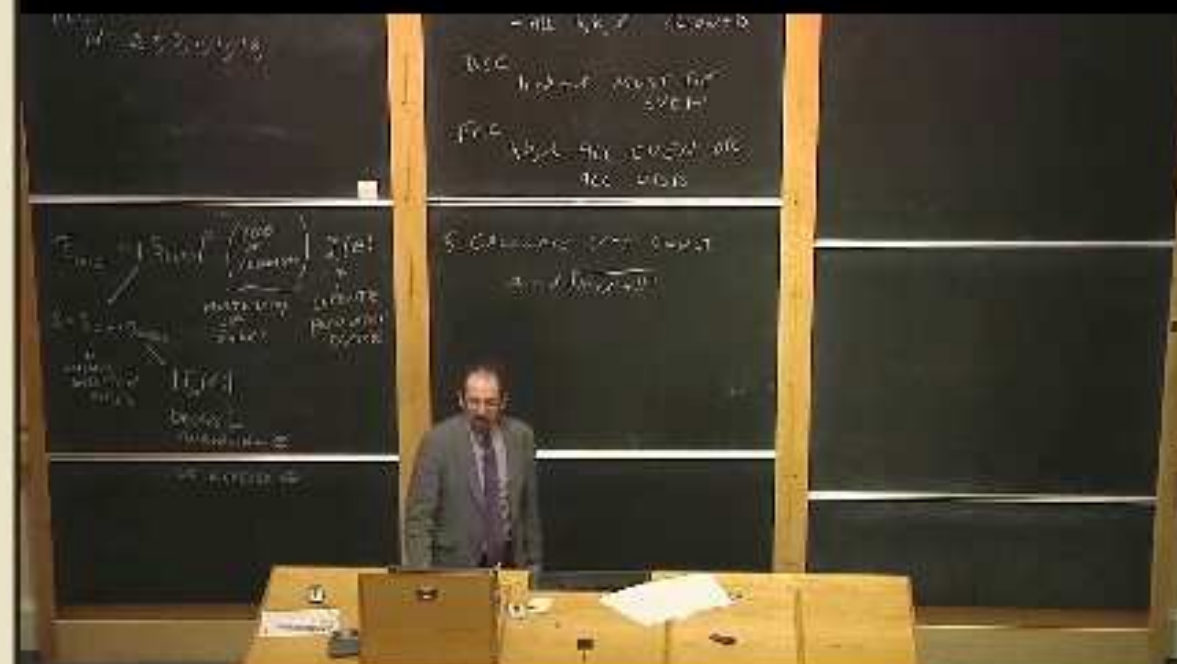
Sidhu et al, J. Applied Physics, 30 1323 (1959).



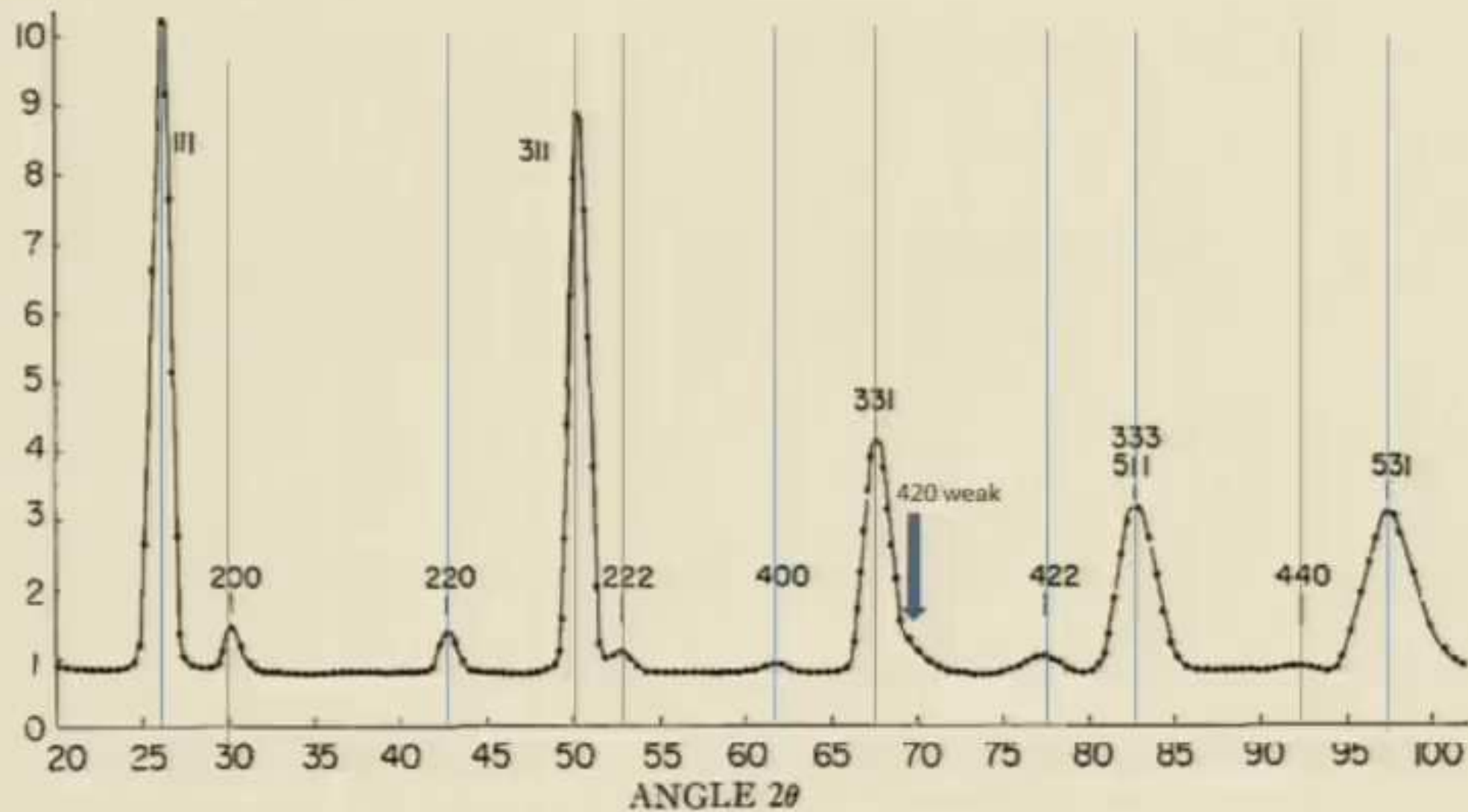
$$a^2/d^2 = h^2 + k^2 + l^2$$

$$d = \frac{\lambda}{2 \sin \theta}$$

Peak	Angle 2θ	
a	26	2.42Å
b	30.1	2.10Å
c	42.8	1.49Å
d	50.2	1.28Å
e	52.8	1.23Å
f	62.4	1.05Å
g	67.6	0.98Å
h	70	0.95Å

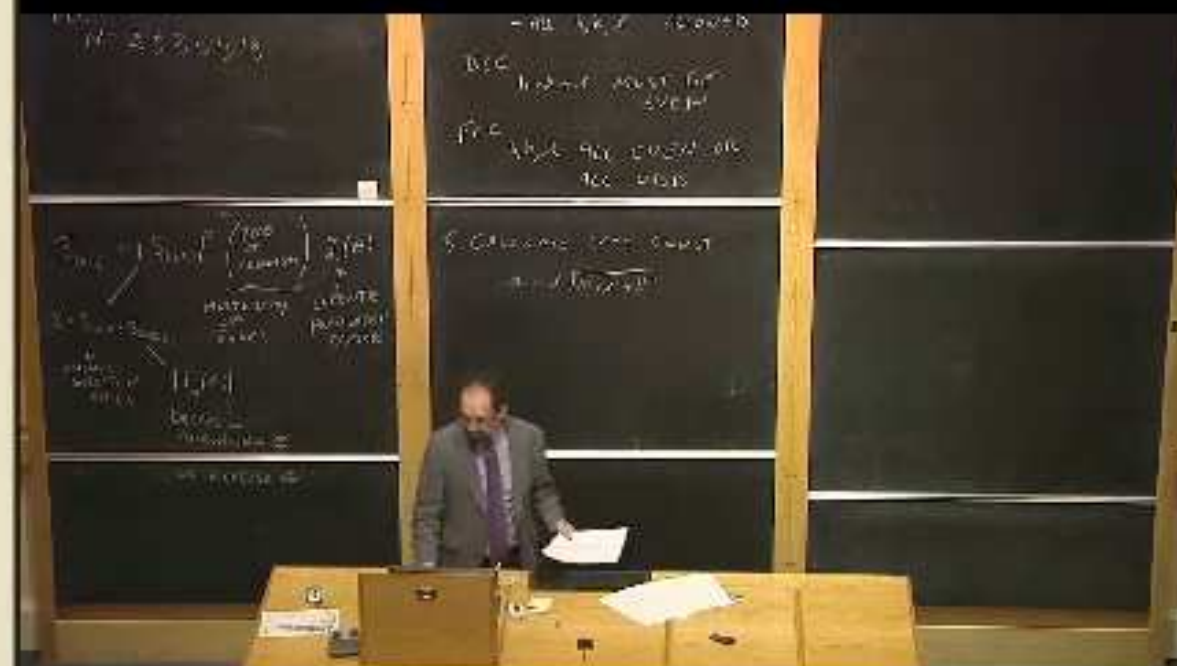


$\lambda = 1.09$ Angstrom TiC



Multiplicity: 8 6 12 24 8 6 24 24 24 24+6 24 24

can we figure out what the unit cell looks like?



NaCl structure

Ti @ [0,0,0]

C @ [1/2,1/2,1/2]

$$|S|^2 = |b_{Ti} + b_C(-1)^{h+k+l}|^2$$

$$= |b_{Ti} - b_C|^2 \text{ for } h+k+l \text{ odd}$$

$$= |b_{Ti} + b_C|^2 \text{ for } h+k+l \text{ even}$$

ZnS structure

Ti @ [0,0,0]

C @ [1/4,1/4,1/4]

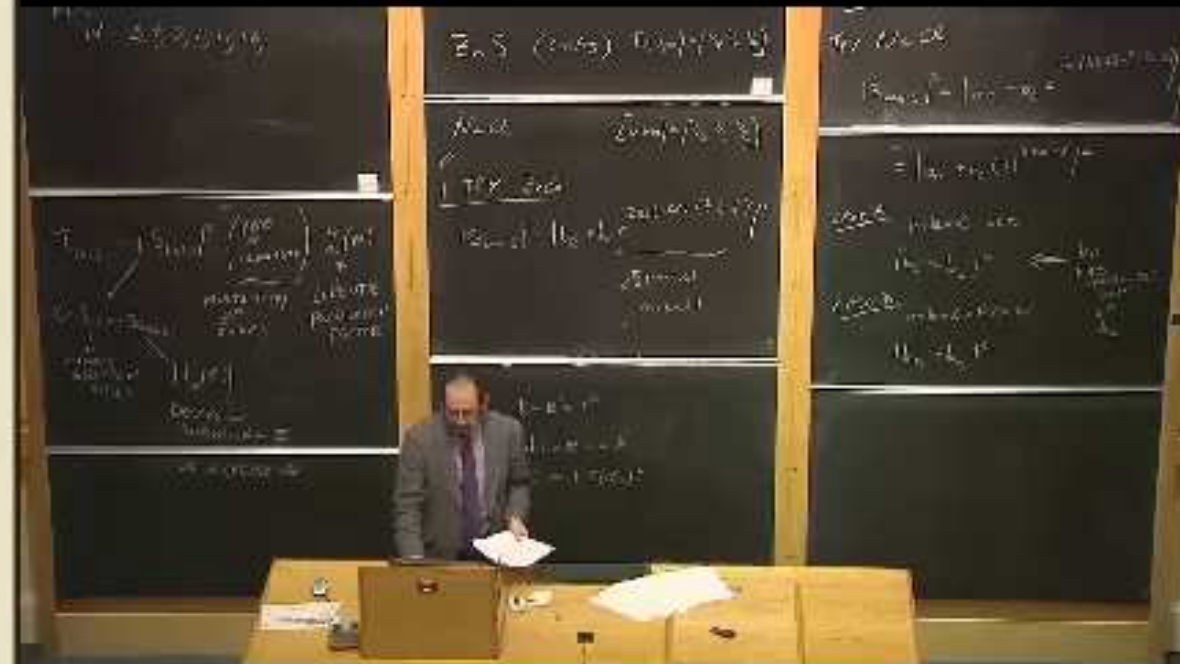
$$|S|^2 = |b_{Ti} + b_C(i)^{h+k+l}|^2$$

$$= b_{Ti}^2 + b_C^2 \text{ for } h+k+l \text{ odd}$$

$$= |b_{Ti} + b_C|^2 \text{ for } h+k+l = 4m$$

$$= |b_{Ti} - b_C|^2 \text{ for } h+k+l = 4m+2$$

Conclude must be NaCl structure



The Rutherford-Appleton Lab in Oxfordshire



Spallation Neutron Source

Synchrotron X-ray Source

